

LOW ACRYLONITRILE NBR PATENT LANDSCAPE: SYNTHESIS, COMPOSITION CONTROL, AND APPLICATION-DRIVEN FORMULATION TRENDS

1. ABSTRACT

This report synthesizes four primary patents relevant to nitrile butadiene rubber (NBR) with emphasis on low-acrylonitrile embodiments and adjacent NBR manufacturing/formulation strategies. The selected landscape spans one electro-conductive member patent that explicitly uses NBR as a matrix/domain rubber system, and three NBR composition patents focused on hydrogenated NBR, NBR/PVC blends, and NBR-based rubber-metal laminates. Although not all patents are direct low-ACN synthesis disclosures, each contributes evidence on how acrylonitrile content is selected, constrained, or leveraged in NBR materials design.

Across the set, the most explicit acrylonitrile-content disclosures appear in the hydrogenated NBR composition patent, which defines hydrogenated NBRs at 25–44 wt% acrylonitrile content and further narrows preferred embodiments to 30–40 wt%. The electro-conductive member patent discloses NBR as one of the paired diene rubbers in a matrix/domain architecture, with examples listing commercial NBR grades spanning 15.0–41.5 wt% nitrile content, including a low-nitrile grade at 15%. The NBR/PVC sealing rubber patent centers on a polymer blend containing NBR and PVC, with a PVC content window of 15.70–17.50 wt% and examples using a high-nitrile NBR polymer at 42.5 wt% nitrile content. The rubber-metal laminate patent uses an NBR polymer with 34 wt% nitrile content and emphasizes carbon-black architecture and solvent-based coating formulation rather than monomer synthesis.

From a process perspective, the patents are dominated by compounding, kneading, vulcanization, and laminate/coating operations rather than direct NBR polymerization. The only synthesis-adjacent NBR identity control is through selection of commercial NBR grades or blends with specified nitrile content. Major formulation trends include sulfur or peroxide curing, use of plasticizers, carbon black/silica reinforcement, and careful tuning of filler type and amount to balance oil resistance, compression set, surface smoothness, and mechanical recovery. Water is not a polymerization medium in these disclosures; when present, it is only implicit or absent from the extracted evidence. Overall, the evidence supports low-ACN NBR as a composition-selection problem more than a new polymerization route problem in the retrieved patents.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US11347156B2
- Patent Title: Electro-conductive member for electrophotography, process cartridge for ...
- Assignee: Canon Kabushiki Kaisha
- Jurisdiction: US
- Publication Year: 2022
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent does not disclose NBR polymerization; instead, it discloses an electro-conductive member built from a matrix/domain rubber architecture. The NBR appears as one of the diene rubbers used in the matrix or domain compositions, with performance controlled by rubber pairing, solubility-parameter difference, viscoelastic balance, and conductive-particle loading. - Acrylonitrile content: 15% — NBR (1), JSR NBR N260S - Acrylonitrile content: 41.5% — NBR (2), JSR NBR N220S - Acrylonitrile content: 27.5% — NBR (3), Nipol DN302 - Acrylonitrile content: 18.0% — NBR (4), Nipol DN401LL - Acrylonitrile content: 35% — NBR (5), Nipol N230S - Acrylonitrile content: 40.0% — NBR (6), JSR NBR N202S - NBR composition for domain: 100 parts by mass — Example 1, Table 1 - Carbon black: 60 parts by mass — Example 1, Table 1 - Zinc oxide: 5 parts by mass — Example 1, Table 1 - Zinc stearate: 2 parts by mass — Example 1, Table 1 - Polyisoprene: 70 parts by mass — Example 1, Table 2 - Sulfur: 3 parts by mass — Example 1, Table 3 - Di-2-benzothiazolyl disulfide: 2 parts by mass — Example 1, Table 3 - Tetramethylthiuram disulfide: 0.5 parts by mass — Example 1, Table 3 - Temperature: 23° C. — claim 1 / viscoelasticity definition - Relative humidity: 50% — claim 1 / viscoelasticity definition - Frequency: 80 Hz — claim 1 / viscoelasticity definition - Difference of absolute SP values: 0.2 to 4.0 (J/cm³)^{0.5} — claim 1 - $\tan \delta_1 / \tan \delta_2$: 0.45 to 2.00 — claim 1 - Electro-conductive particles in domains: 50 to 150 parts by mass per 100 parts by mass of second rubber — claim 1 - Domain volume fraction: 26.4 vol % — Comparative Example 1, Table 8 - Domain volume fraction: 25.9 vol % — Comparative Example 2, Table 8 - Domain volume fraction: 28.1 vol % — Comparative Example 3, Table 8 - Domain volume fraction: 29.9 vol % — Comparative Example 4, Table 8 - Domain volume fraction: 21.8 vol % — Comparative Example 5, Table 8 - Difference between SP values: 0.5 — Comparative Example 1, Table 8 - Difference between SP values: 2.1 — Comparative Example 2, Table 8 - Difference between SP values: 0 — Comparative Example 3, Table 8 - Difference between SP values: 4.2 — Comparative Example 4, Table 8 - Difference between SP values: 0.6 — Comparative Example 5, Table 8 - Difference between SP values: 4.0 — Comparative Example 6, Table 8 - Volume resistivity (matrix): 9.70E-06 $\Omega \cdot \text{cm}$ — Comparative Example 1, Table 8 - Volume resistivity (matrix):

8.10E-10 $\Omega \cdot \text{cm}$ — Comparative Example 3, Table 8 - Volume resistivity (matrix): 1.90E-15 $\Omega \cdot \text{cm}$ — Comparative Example 4, Table 8 - Volume resistivity (matrix): 2.90E-11 $\Omega \cdot \text{cm}$ — Comparative Example 5, Table 8 - Domain size: 3.2 μm — Comparative Example 1, Table 8 - Domain size: 4.5 μm — Comparative Example 3, Table 8 - Domain size: 0.7 μm — Comparative Example 4, Table 8 - Domain size: 2.5 μm — Comparative Example 5, Table 8 - Distance between domains: 2.0 μm — Comparative Example 1, Table 8 - Distance between domains: 3.1 μm — Comparative Example 3, Table 8 - Distance between domains: 0.3 μm — Comparative Example 4, Table 8 - Distance between domains: 1.7 μm — Comparative Example 5, Table 8 - MD-1 hardness: 60° — Comparative Example 1, Table 8 - MD-1 hardness: 58° — Comparative Example 2, Table 8 - MD-1 hardness: 61° — Comparative Example 3, Table 8 - MD-1 hardness: 51° — Comparative Example 4, Table 8 - MD-1 hardness: 82° — Comparative Example 5, Table 8 - MD-1 hardness: 59° — Comparative Example 6, Table 8 - Medium & Water Role: No water-related process step disclosed in the extracted evidence. - acrylonitrile content: 15% to 41.5% nitrile content in listed NBR starting materials; low-nitrile grade NBR (1) at 15% and NBR (4) at 18.0%

Medium & Water Role - No water-related process step disclosed in the extracted evidence. - Core reaction medium: No polymerization medium disclosed; rubber compounding and molding in solid-state/processing media only

acrylonitrile content - 15% to 41.5% nitrile content in listed NBR starting materials; low-nitrile grade NBR (1) at 15% and NBR (4) at 18.0%

Relevant Experimental Evidence - Example 1: NBR was used as the raw rubber in the domain composition, with a 100-part formulation and conductive filler package. - Comparative Example 3: NBR was used in the matrix while EPDM was used in the domain, showing poor matrix-domain performance despite NBR presence. - Comparative Example 5: NBR was used in both domain and matrix, but the matrix-domain structure could not be confirmed and only NBR was detected. - Table 8: NBR grades spanning 15% to 41.5% nitrile content were listed as starting materials, including a low-nitrile grade at 15%. - Example 40: the electro-conductive member was used as a transfer member and evaluated under high-speed printing conditions.

Technical Relevance This patent is relevant because it explicitly includes NBR among the diene rubbers used in a performance-controlled rubber architecture and discloses commercial NBR grades with low nitrile content. While not a synthesis patent, it provides composition-selection evidence for low-ACN NBR grades in functional rubber systems.

Patent 2

- Patent Number: US7772305B2
- Patent Title: Hydrogenated NBR composition
- Assignee: Nok Corporation
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown

- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a hydrogenated NBR composition rather than NBR polymerization. The composition is built from hydrogenated NBR, ester-based plasticizer, and organic peroxide, with optional polyfunctional unsaturated compound and fillers, followed by kneading and press/oven vulcanization. - Acrylonitrile content: 25-44% by weight — claim 1 / abstract - Iodine number: 32-65 — abstract - Acrylonitrile content: 25-44% by weight — claim 1 - Iodine number: 40-65 — claim 1 - Acrylonitrile content: 30-40% by weight — claim 2 - Iodine number: 40-65 — claim 2 - Di(butoxyethoxyethyl) adipate plasticizer: 2-23 parts by weight — claim 1 - Organic peroxide: 0.5-10 parts by weight — claim 1 - Polyfunctional unsaturated compound: 0.5-10 parts by weight — claim 3 - Carbon black / white carbon / mixture: 20-150 parts by weight — claim 4 - Hydrogenated NBR (Zetpole 2020): AN content 36 wt. % — Comparative Example 1 - Hydrogenated NBR (Zetpole 2020): iodine number 28 — Comparative Example 1 - FEF carbon black: 32 parts by weight — Comparative Example 1 - MT carbon black: 35 parts by weight — Comparative Example 1 - Ester-based plasticizer (RS107): 15 parts by weight — Comparative Example 1 - Antioxidant (Anti CD): 1.5 parts by weight — Comparative Example 1 - Antioxidant (Anti MBZ): 1.5 parts by weight — Comparative Example 1 - Zinc white: 2 parts by weight — Comparative Example 1 - Organic peroxide (Percumyl D): 6 parts by weight — Comparative Example 1 - Press vulcanization: 170° C. for 20 minutes — Comparative Example 1 - Oven vulcanization: 160° C. for 3 hours — Comparative Example 1 - Hydrogenated NBR (Zetpole 2000): AN content 36 wt. % — Comparative Example 2 - Hydrogenated NBR (Zetpole 2000): iodine number 4 — Comparative Example 2 - Hydrogenated NBR (Zetpole 2010): AN content 36 wt. % — Comparative Example 3 - Hydrogenated NBR (Zetpole 2010): iodine value 11 — Comparative Example 3 - Hydrogenated NBR (Zetpole 1020): AN content 44 wt. % — Comparative Example 4 - Hydrogenated NBR (Zetpole 1020): iodine number 25 — Comparative Example 4 - Hydrogenated NBR (Zetpole 3120): AN content 25 wt. % — Comparative Example 5 - Hydrogenated NBR (Zetpole 3120): iodine number 31 — Comparative Example 5 - Hydrogenated NBR (Zetpole 2030L): AN content 36 wt. % — Example 1 - Hydrogenated NBR (Zetpole 2030L): iodine number 57 — Example 1 - Blend of Zetpole 2020 and Zetpole 2030L: AN content 36 wt. % — Example 2 - Blend of Zetpole 2020 and Zetpole 2030L: iodine number 34 — Example 2 - Blend of Zetpole 2020 and Zetpole 2030L: 80 wt. % / 20 wt. % — Example 2 - Blend of Zetpole 2020 and Zetpole 2030L: AN content 36 wt. % — Example 3 - Blend of Zetpole 2020 and Zetpole 2030L: iodine number 40 — Example 3 - Blend of Zetpole 2020 and Zetpole 2030L: 60 wt. % / 40 wt. % — Example 3 - Blend of Zetpole 2020 and Zetpole 2030L: AN content 36 wt. % — Example 4 - Blend of Zetpole 2020 and Zetpole 2030L: iodine number 48 — Example 4 - Blend of Zetpole 2020 and Zetpole 2030L: 30 wt. % / 70 wt. % — Example 4 - Ester-based plasticizer: 5 parts by weight — Example 5 - Ester-based plasticizer: 10 parts by weight — Example 6 - Ester-based plasticizer: 20 parts by weight — Example 7 - FEF carbon black: 15 parts by weight — Example 8 - White carbon (Nipsil E74P): 25 parts by weight — Example 8 - Silane coupling agent (A172): 2 parts by weight — Example 8 - White carbon (Nipsil E74P): 43 parts by weight — Example 9 - Silane coupling agent (A172): 2 parts by weight — Example 9 - Medium & Water Role: No water-related process step disclosed in the extracted evidence. - acrylonitrile content: 25-44% by weight; preferred 30-40% by weight; examples include

25 wt.%, 36 wt.%, and 44 wt.% AN content

Medium & Water Role - No water-related process step disclosed in the extracted evidence. - Core reaction medium: No polymerization medium disclosed; compounding and vulcanization in solid-state processing

acrylonitrile content - 25-44% by weight; preferred 30-40% by weight; examples include 25 wt.%, 36 wt.%, and 44 wt.% AN content

Relevant Experimental Evidence - Example 1: a hydrogenated NBR with 36 wt.% AN content and iodine number 57 was used in a peroxide-cured composition. - Examples 2-4: blends of hydrogenated NBRs were used to keep AN content at 36 wt.% while varying iodine number from 34 to 48. - Examples 5-7: ester-based plasticizer level was varied from 5 to 20 parts by weight to study property balance. - Examples 8-9: carbon black was partially or fully replaced with white carbon plus silane coupling agent to evaluate reinforcement effects. - Comparative Example 5: a hydrogenated NBR with 25 wt.% AN content and iodine number 31 was tested as a lower-AN reference.

Technical Relevance This patent is highly relevant because it explicitly defines the acrylonitrile-content window for hydrogenated NBR compositions and includes a lower-end 25 wt.% AN embodiment. Although hydrogenated, it provides a clear composition benchmark for low-ACN nitrile elastomer selection and downstream curing/formulation.

Patent 3

- Patent Number: US8846804B2
- Patent Title: NBR composition and sealing rubber material
- Assignee: Nok Corporation
- Jurisdiction: US
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses an NBR/PVC-based sealing rubber material prepared by kneading an NBR polymer with PVC and compounding sulfur, accelerators, fillers, and processing aids, followed by primary and secondary vulcanization. The emphasis is on formulation control, especially PVC content and suppression of wax/stearic-acid-derived insolubles. - PVC content: 15.70 to 17.50 wt % — abstract / claim 1 - First compression set: 46% or less — claim 2 - Second compression set: 46% or less — claim 2 - High nitrile (NBR) polymer nitrile content: 42.5% — Example 1 - High nitrile (NBR) polymer Mooney viscosity [ML1+4 (100° C.)]: 77.5 — Example 1 - NBR/PVC blend polymer PVC content: 35% — Example 1 - SRF class carbon black: 70 parts by weight — Example 1 - Zinc oxide: 5 parts by weight — Example 1 - Age resister: 2 parts by weight — Example 1 - Plasticizer (Ethyl

hexyl adipate): 30 parts by weight — Example 1 - Sulfur: 0.5 part by weight — Example 1 - CBS: 2 parts by weight — Example 1 - TMTD: 2.5 parts by weight — Example 1 - Kneader: 3 L — Example 1 - Primary vulcanization: 160 to 180° C. for 10 to 20 minutes — Example 1 - Secondary vulcanization: 160° C. for 1 hour — Example 1 - Fuel C immersion: 20 g rubber pieces in 50 ml — Example 1 - Fuel C composition: toluene/isooctane = 50:50 (volume ratio) — Example 1 - Immersion temperature: 40° C. for 5 days — Example 1 - Concentration step: 10 ml — Example 1 - Low-temperature precipitation step: -15° C. for 5 hours — Example 1 - Centrifugation: 2,000 rpm for 5 minutes — Example 1 - O-ring size: G25 — Example 1 - Compression ratio: 25% — Example 1 - Test temperature: 100° C. for 70 hours — Example 1 - Second compression set test: 100° C. for 15 hours — Example 1 - Comparative Example 1: 100 parts by weight of high nitrile polymer — Comparative Example 1 - Comparative Example 2: 60 parts by weight of high nitrile polymer and 40 parts by weight of NBR/PVC blend polymer — Comparative Example 2 - Comparative Example 3: 45 parts by weight of high nitrile polymer and 55 parts by weight of NBR/PVC blend polymer — Comparative Example 3 - Comparative Example 4: 1 part by weight stearic acid and 3 parts by weight wax — Comparative Example 4 - Comparative Example 5: 1 part by weight stearic acid — Comparative Example 5 - Comparative Example 6: 1 part by weight wax — Comparative Example 6 - Comparative Example 7: 0.5 part by weight stearic acid and 0.5 wt % wax — Comparative Example 7 - Comparative Example 8: 0.1 part by weight stearic acid and 0.1 part by weight wax — Comparative Example 8 - Medium & Water Role: No water-related process step disclosed in the extracted evidence. - acrylonitrile content: Not disclosed in extracted evidence

Medium & Water Role - No water-related process step disclosed in the extracted evidence. - Core reaction medium: Mixed-solvent coating solution after kneading; solid-state compounding and oven vulcanization

acrylonitrile content - Not disclosed in extracted evidence

Relevant Experimental Evidence - Example 1: an NBR/PVC blend polymer with 35% PVC content was compounded with sulfur, CBS, and TMTD and vulcanized into a sealing material. - Comparative Examples 1-3: changing the ratio of high nitrile polymer to NBR/PVC blend polymer was used to probe the effect of polymer blend balance. - Comparative Examples 4-8: stearic acid and wax levels were varied to assess precipitation/insoluble-substance formation and processability. - The process used a 3 L pressure kneader followed by open-roll addition of accelerators and staged vulcanization. - The patent explicitly ties PVC content control to ozone resistance, ejectability from kneader, and compression-set performance.

Technical Relevance This patent is relevant because it is an NBR-based sealing composition patent that explicitly uses an NBR polymer and defines a composition window for the NBR/PVC blend. While not a synthesis route, it provides formulation evidence for low-ACN NBR-related sealing materials and their performance constraints.

Patent 4

- Patent Number: US9688834B2
- Patent Title: NBR composition and rubber-metal laminate

- Assignee: Nok Corporation
- Jurisdiction: US
- Publication Year: 2017
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses an NBR composition for rubber-metal laminates, prepared by kneading NBR with carbon blacks, sulfur, accelerators, silica, zinc oxide, stearic acid, and age resistor, then dissolving the compound in a toluene/methyl ethyl ketone mixed solvent to form a coating solution and vulcanizing it in heated air. The focus is on balancing reinforcement, surface smoothness, and adhesion rather than polymer synthesis. - Acrylonitrile content: 34% — NBR (N237) in Example 1 - MT carbon black: 150 parts by weight — Example 1 - Carbon black (SEAST TA): 40 parts by weight — Example 1 - Sulfur: 2 parts by weight — Example 1 - Tetrabenzylthiuram disulfide: 5 parts by weight — Example 1 - N-cyclohexyl-2-benzothiazolyl sulfonamide: 9 parts by weight — Example 1 - Silica: 30 parts by weight — Example 1 - Zinc oxide: 5 parts by weight — Example 1 - Stearic acid: 2 parts by weight — Example 1 - Age resistor: 5 parts by weight — Example 1 - Toluene to methyl ethyl ketone ratio: 9:1 — Example 1 - Solid concentration: 25% by weight — Example 1 - Dried thickness: 100 μm — Example 1 - Vulcanization temperature: 210° C. — Example 1 - Vulcanization time: 3 minutes — Example 1 - Contact pressure: 200 MPa — Example 1 - Press temperature: 150° C. — Example 1 - Abrasion resistance criterion: less than 0.2 cc — Example 1 - Surface roughness criterion Ra: less than 0.3 μm — Example 1 - Surface roughness criterion Ry: less than 2.2 μm — Example 1 - NBR polymer: 100 parts by weight — Example 4/6 tables - MT carbon black: 90, 150, 250, 50, 210 parts by weight — Example 4/6 tables - SEAST TA carbon black: 70, 0, 100, 10, 70 parts by weight — Example 4/6 tables - Tetrabenzylthiuram disulfide: 5 parts by weight — Example 4/6 tables - N-cyclohexyl-2-benzothiazolyl sulfenamide: 9, 5, 12 parts by weight — Example 4/6 tables - Silica: 30 parts by weight — Example 4/6 tables - Zinc oxide: 5 parts by weight — Example 4/6 tables - Stearic acid: 2 parts by weight — Example 4/6 tables - Age resistor: 5 parts by weight — Example 4/6 tables - Medium & Water Role: No water-related process step disclosed in the extracted evidence. - acrylonitrile content: 34% nitrile content — NBR (N237) used in Example 1

Medium & Water Role - No water-related process step disclosed in the extracted evidence. - Core reaction medium: Organic solvent coating solution (toluene/methyl ethyl ketone 9:1) after solid-state kneading

acrylonitrile content - 34% nitrile content — NBR (N237) used in Example 1

Relevant Experimental Evidence - Example 1: an NBR with 34% nitrile content was compounded with a dual-carbon-black system and dissolved in a toluene/MEK solvent to make a coating solution. - Example 1: the coating was vulcanized at 210° C. for 3 minutes to form a rubber-metal laminate. - Example 4: varying the ratio of MT carbon black to SEAST TA changed flow resistance, abrasion resistance, surface roughness, and adhesion. - Example 6: a higher MT

carbon black loading of 210 parts by weight with 10 parts SEAST TA was evaluated against other filler balances. - The patent demonstrates that carbon-black architecture strongly controls laminate surface quality and reinforcement.

Technical Relevance This patent is relevant because it explicitly discloses an NBR polymer with 34% nitrile content and shows how NBR formulation is tuned for laminate performance. It supports the broader low-to-mid acrylonitrile NBR formulation space, even though it is not a polymerization patent.

PATENT COMPARISON TABLE

Patent	Applicant	Priority	Medium & Water Role	acrylonitrile content	Key Findings
US11347156B2	Canon Kabushiki Kaisha	2022	No water-related process step disclosed in the extracted evidence.	15% to 41.5% nitrile content in listed NBR starting materials; low-nitrile grade NBR (1) at 15% and NBR (4) at 18.0%	Example 1: NBR was used as the raw rubber in the domain composition, with a 100-part formulation and conductive filler p
US7772305B2	Nok Corporation	2010	No water-related process step disclosed in the extracted evidence.	25-44% by weight; preferred 30-40% by weight; examples include 25 wt.%, 36 wt.%, and 44 wt.% AN content	Example 1: a hydrogenated NBR with 36 wt.% AN content and iodine number 57 was used in a peroxide-cured composition.

US8846804B2	Nok Corporation	2014	No water-related process step disclosed in the extracted evidence.	Not disclosed in extracted evidence	Example 1: an NBR/PVC blend polymer with 35% PVC content was compounded with sulfur, CBS, and TMTD and vulcanized into a
US9688834B2	Nok Corporation	2017	No water-related process step disclosed in the extracted evidence.	34% nitrile content — NBR (N237) used in Example 1	Example 1: an NBR with 34% nitrile content was compounded with a dual-carbon-black system and dissolved in a toluene/MEK

SUPPORTING / RELATED PATENTS (SECONDARY)

Supporting Patent 1

- Patent Number: US7592409B2
- Patent Title: Styrene acrylonitrile isoprene triblock copolymer
- Assignee: The Goodyear Tire & Rubber Company
- Jurisdiction: US
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: The patent is about a styrene acrylonitrile isoprene triblock copolymer, which is a different block polymer architecture and not NBR itself.

Supporting Patent 2

- Patent Number: US12371558B2
- Patent Title: Resin-extended rubber and process for preparing
- Assignee: Bridgestone Americas Tire Operations, Llc
- Jurisdiction: US
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: The patent is about resin-extended rubber compositions and solution synthesis of styrene-butadiene rubber, not NBR.

Supporting Patent 3

- Patent Number: US12286702B2
- Patent Title: Method for the production of functionalized elastomeric manufactured articles ...
- Assignee: Wise S.R.L.
- Jurisdiction: US
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: The patent concerns functionalized elastomeric manufactured articles and only lists NBR among many possible support materials.

Supporting Patent 4

- Patent Number: US20230374277A1
- Patent Title: Composite electrolytes with binders
- Assignee: Blue Current, Inc.
- Jurisdiction: US
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: This patent is about composite electrolytes and binders, where NBR-like materials appear only as binder options rather than the central invention.

Supporting Patent 5

- Patent Number: EP3902112B1
- Patent Title: Adhesively laminated core for stator, method of manufacturing the same, and ...
- Assignee: Nippon Steel Corporation
- Jurisdiction: EP
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: The invention is primarily an adhesively laminated stator core and adhesive system, with NBR only as an elastomer additive in the adhesive.

Supporting Patent 6

- Patent Number: US3297792A

- Patent Title: Blends of (1) graft copolymer of vinyl chloride onto chlorinated polyolefin, (2 ...
- Assignee: Monsanto Co
- Jurisdiction: US
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: The patent is a multicomponent blend where the acrylonitrile-diene rubber is only one ingredient, so it is not primarily about NBR synthesis or manufacture.

Supporting Patent 7

- Patent Number: US8022136B2
- Patent Title: Vulcanized rubber composition, pneumatic tire and the process of producing the ...
- Assignee: Sumitomo Rubber Industries, Ltd.
- Jurisdiction: US
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: This is a vulcanized rubber composition and tire patent where NBR is only one possible rubber component in a blend.

Supporting Patent 8

- Patent Number: US9376537B2
- Patent Title: Process for preparing completely delaminated graphene oxide/rubber ...
- Assignee: Beijing University Of Chemical Technology
- Jurisdiction: US
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: The patent is about graphene oxide/rubber composites, and NBR is only listed as one possible latex component.

Supporting Patent 9

- Patent Number: US9328221B2
- Patent Title: Method for preparing graphene oxide/white carbon black/rubber nanocomposite ...
- Assignee: Beijing University Of Chemical Technology
- Jurisdiction: US
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: The patent is about graphene oxide/white carbon black/rubber nanocomposites, not NBR synthesis itself.

Supporting Patent 10

- Patent Number: US9748016B2
- Patent Title: Process for producing carbon nanotube composition and carbon nanotube ...

- Assignee: Zeon Corporation
- Jurisdiction: US
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: The patent is about carbon nanotube compositions, and NBR appears only as one example elastomer among many.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US7772305B2 is the only patent that explicitly defines an acrylonitrile-content window for the target material, at 25–44 wt% AN, with a preferred 30–40 wt% range; its examples center on 36 wt% AN hydrogenated NBR.
- US9688834B2 discloses a conventional NBR polymer at 34% nitrile content in a laminate formulation, placing it within the low-to-mid ACN range but without a synthesis claim.
- US11347156B2 lists multiple commercial NBR grades spanning 15% to 41.5% nitrile content, including a low-nitrile 15% grade, but uses them as functional rubbers in a matrix/domain electro-conductive member rather than as a synthesis target.
- US8846804B2 uses a high nitrile NBR polymer at 42.5% nitrile content and an NBR/PVC blend polymer; it is composition-focused and does not disclose a low-ACN synthesis route.
- All four patents are compounding- and performance-driven rather than monomer-feed polymerization disclosures; none provide direct emulsion or solution copolymerization conditions for making low-ACN NBR.
- Water is absent from the extracted process evidence for all four patents; the disclosed operations are kneading, solvent dissolution, molding, coating, and vulcanization.
- The strongest low-ACN-relevant numerical evidence is the 15% nitrile NBR grade in US11347156B2 and the 34% nitrile NBR in US9688834B2, while US7772305B2 provides the clearest target window for hydrogenated NBR compositions.

4. CONCLUSION

The retrieved patents indicate that low-acrylonitrile NBR is treated primarily as a composition-selection and formulation problem rather than a newly disclosed polymerization process. The most directly useful target window in the set is the 25–44 wt% acrylonitrile range in US7772305B2, with practical examples around 36 wt% AN, while US11347156B2 and US9688834B2 demonstrate that commercial NBR grades at 15–34% nitrile content are actively used in functional rubber systems. For synthesis-oriented work, the evidence does not provide direct copolymerization conditions; therefore, recommended parameters must be inferred cautiously from the disclosed material selections: use NBR grades in the low-to-mid ACN range, control filler and plasticizer balance, and select curing systems matched to the intended application. The patents collectively support sulfur or peroxide curing, careful carbon-black

architecture, and solvent-based or solid-state compounding as the dominant downstream processing strategies for low-ACN NBR materials.

5. REFERENCES

- US11347156B2 | Electro-conductive member for electrophotography, process cartridge for ... | Canon Kabushiki Kaisha | US | 2022 | <https://patents.google.com/patent/US11347156B2/en>
- US7772305B2 | Hydrogenated NBR composition | Nok Corporation | US | 2010 | <https://patents.google.com/patent/US7772305B2/en>
- US8846804B2 | NBR composition and sealing rubber material | Nok Corporation | US | 2014 | <https://patents.google.com/patent/US8846804B2/en>
- US9688834B2 | NBR composition and rubber-metal laminate | Nok Corporation | US | 2017 | <https://patents.google.com/patent/US9688834B2/en>